

Exposure to untrustworthy websites in the 2020 US election

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Analysis of 2020 untrustworthy website exposure using NewsGuard's Trust Score

The NewsGuard database allows us to conduct an additional comparison between the percentage of the population exposed to untrustworthy websites in 2020 and in 2016. In addition to rating sites on a binary criteria of whether or not they repeatedly publish false content, NewsGuard also provides a continuous “trust score” for each website they evaluate, ranging from 0 to 100 with 0 being the lowest credibility and 100 being the highest. These scores are based on a weighted sum of 9 different criteria of journalistic integrity (one of which being whether or not a site repeatedly publishes false content). Examples of sites with a score of 0 include fawkes-news.com and dmaga2020.xyz, sites with a score of 25 include oann.com and infowars.com, Breitbart.com is an example of a site with a score of 50, sites with a score of 75 include nypost.com and thesun.co.uk, and sites with a score of 100 include theguardian.com and theatlantic.com.

We can determine what percentage of our sample was exposed to at least one website below a NewsGuard score threshold, starting at 0 and moving all the way up to 100. In order to achieve the same 44.3% percentage of individuals exposed to at least one untrustworthy website that Guess et al. (2020) found for 2016, we would need to apply a NewsGuard score cutoff of 70 to our 2020 data, meaning that 44.3% of our sample was exposed to at least one site with a NewsGuard score lower than 70. Examples of sites with a NewsGuard score of 70 include msnbc.com and foxnews.com, which have clear partisan leanings (Kim et al., 2022) but are not repeated purveyors of false information and few would likely argue that they represent untrustworthy websites. In other words, to achieve levels of exposure in 2020 similar to those found in 2016, we would have to apply a criteria for untrustworthy websites that is unreasonable and includes several legitimate news outlets.

Probit models for untrustworthy website exposure

Supplementary Table 1

	Exposure to at least one untrustworthy website (binary)	
	2016	2020
Trump supporter	0.428*** (0.058)	0.447*** (0.088)
Political knowledge	0.047** (0.015)	0.180*** (0.041)
Political interest	0.132*** (0.038)	0.153** (0.051)
College	0.103 (0.060)	0.158 (0.093)
Female	-0.062 (0.055)	0.058 (0.086)
Non-white	-0.261*** (0.060)	-0.296** (0.104)
Age 30-44 years	0.095 (0.090)	-0.047 (0.149)
Age 45-64 years	0.161 (0.082)	0.121 (0.136)
Age 65+	0.330*** (0.093)	0.334* (0.145)
Constant	-0.991*** (0.130)	-1.897*** (0.214)
Observations	2,514	1,151
Log Likelihood	-1,405.253	-535.051
Akaike Inf. Crit.	2,830.507	1,090.102

Note:

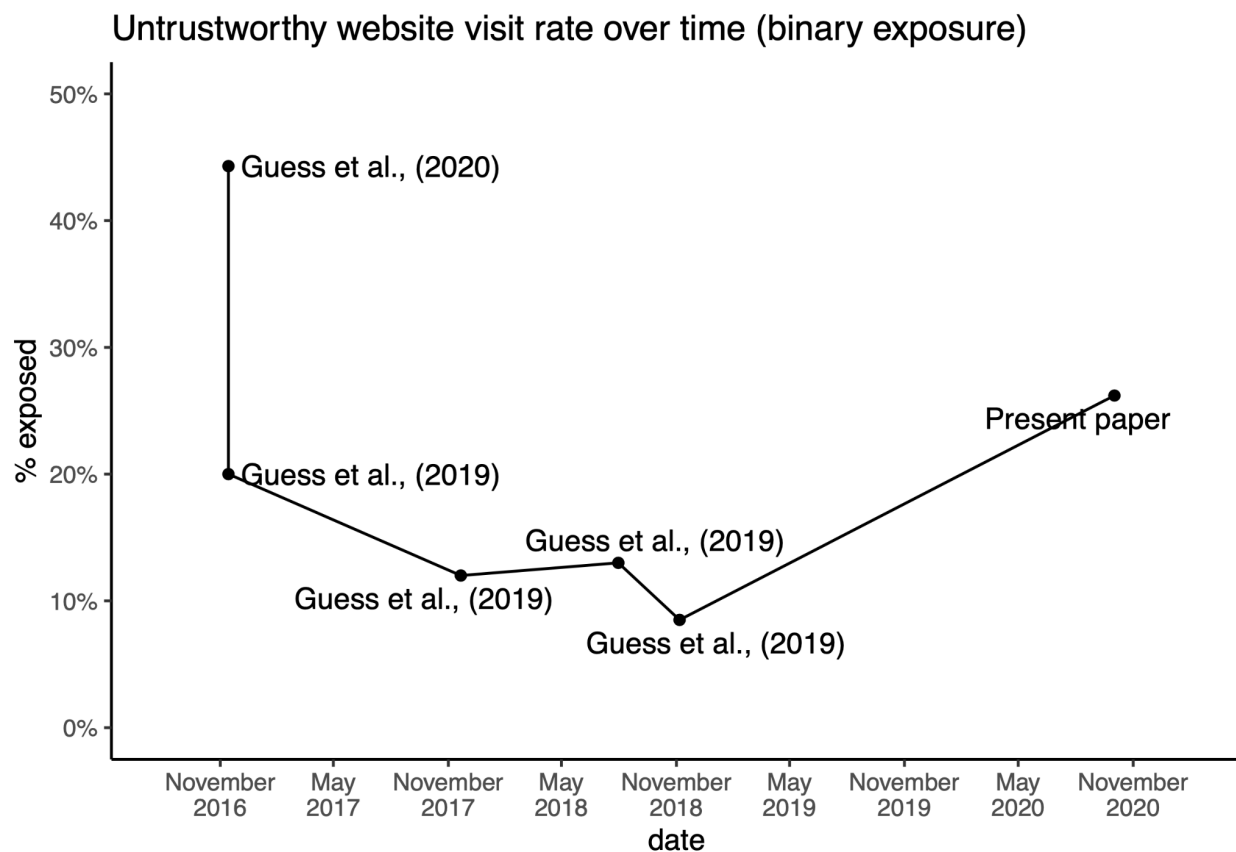
*p<0.05; **p<0.01; ***p<0.001

Supplementary Table 1 Caption: Probit regression coefficients are shown with standard errors in parentheses (models estimated using survey weights). The model on the left represents the 2016 election and the model on the right represents the 2020 election. The dependent variable in both models is a binary variable indicating whether an individual was exposed to at least one untrustworthy website during the data collection period (1) or not (0). “Trump supporter” variable: intending to vote for Trump in 2020 election = 1; not intending to vote for Trump in 2020 election = 0. “Political knowledge” variable: variable ranging from 0-4 representing the number of questions in Pew Research Center’s civic knowledge questionnaire answered correctly out of 4. “Political interest” variable: variable ranging from 1-4 where 4 = people who say they pay attention to what’s going on in government and politics “most of the time” and 1 = those who pay attention “hardly at all”. “College” variable: 1 = college graduate; 0 = not a college graduate. “Female” variable: 1 = indicated identifying as a female, 0 = did not indicate identifying as a female. “Non-white” variable: 1 = indicated identifying as a race other than white; 0 = indicated identifying as white. P-values are two-sided and not corrected for multiple comparisons.

Exposure to untrustworthy websites 2016-2020

Below is a plot of exposure to untrustworthy websites from 2016 - 2020. This plot relies on estimates from Guess et al. (2020) (who have Pulse browsing data for a national sample of Americans from October 7 - November 14, 2016), Guess et al. (2019) (who have Pulse browsing data from October 7 - November 14, 2016, October 25 - November 21, 2017, June 11 - July 31, 2018, and October 5 - November 6, 2018), and the present paper (which has Pulse data from October 2 - November 9, 2020). Estimates are difficult to directly compare across time because of the different untrustworthy website lists used in each study, exemplified by the difference in the estimate for 2016 exposure between Guess et al. (2019) and Guess et al. (2020), which rely on the same web browsing data (see Supplementary Figure 1 caption).

Supplementary Figure 1



Supplementary Figure 1 Caption: List of untrustworthy websites used by Guess et al. (2019) = 42 websites, list of untrustworthy websites used by Guess et al. (2020) = 490 websites, list of untrustworthy websites used in present paper = 1,796 websites.

Exposure to hard news in 2016 and 2020

Below are several analyses of exposure to hard news websites in 2016 compared to 2020. These analyses mirror those of untrustworthy website exposure presented in the main paper. For the purposes of these analyses, hard news websites are those contained in Bakshy et al. (2015) and also all websites rated by NewsGuard, excluding those which NewsGuard determined to repeatedly publish false content (total list of hard news websites = 5,471 websites).

Supplementary Table 2: Percentage of Americans exposed to at least one hard news website

2016		2020		t-value	p-value
M	SE	M	SE		
0.91	0.006	0.897	0.009	-1.265	0.206

Supplementary Table 2 Caption: Weighted two-sample t-test used to calculate p-value.

Supplementary Table 3: Average number of exposures to hard news sites of those exposed

2016		2020		t-value	p-value
M	SE	M	SE		
313.9	26	312.5	28.8	-0.047	0.963

Supplementary Table 3 Caption: Weighted two-sample t-test used to calculate p-value.

The average length of time spent on hard news websites was similar in 2020 (37.9 seconds; 95% CI = 36.8-39.1s) to in 2016 (42.1 seconds; Guess et al. [2020] did not quantify uncertainty for this quantity).

Supplementary Table 4: Percentage of Trump supporters exposed to at least one hard news website

2016		2020		t-value	p-value
M	SE	M	SE		
0.92	0.01	0.91	0.02	-0.354	0.724

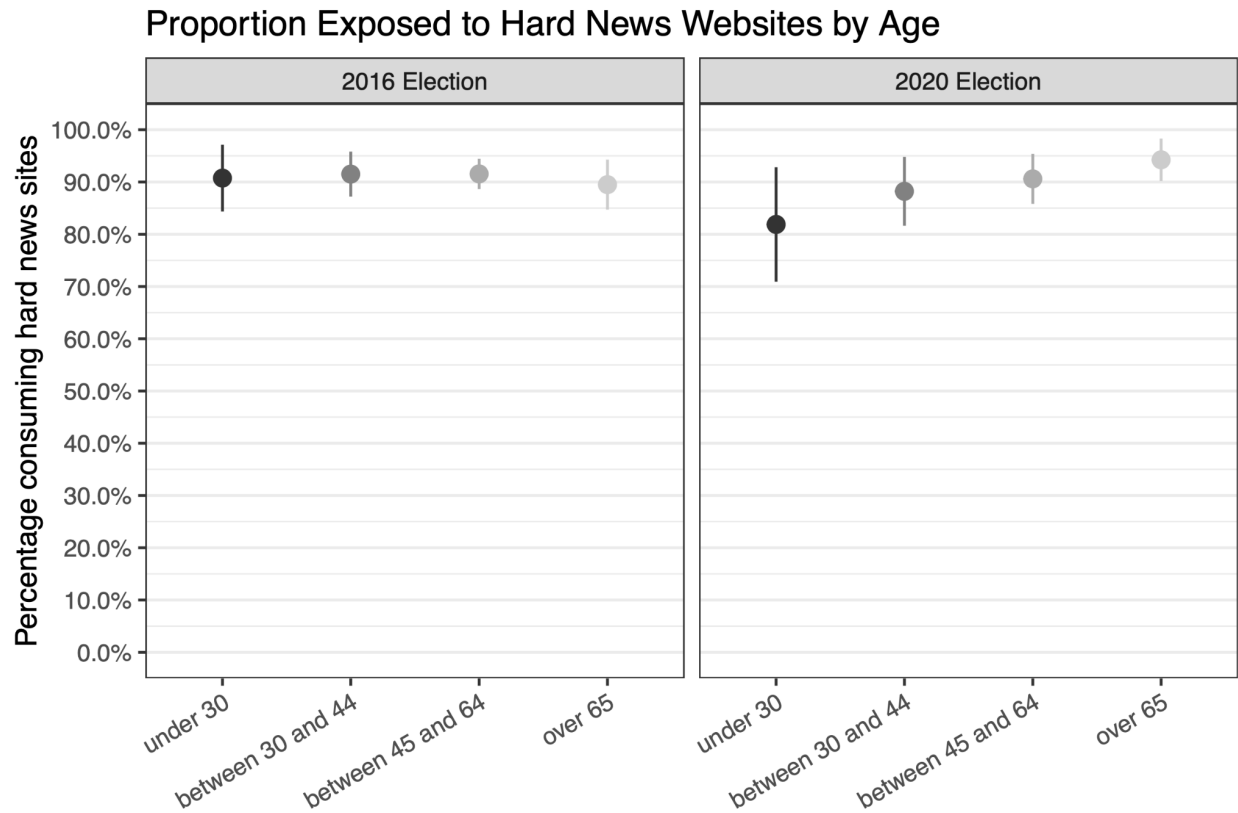
Supplementary Table 4 Caption: Weighted two-sample t-test used to calculate p-value.

Supplementary Table 5: Percentage of Clinton/Biden supporters exposed to at least one hard news website

2016		2020		t-value	p-value
M	SE	M	SE		
0.92	0.01	0.91	0.02	-0.858	0.391

Supplementary Table 5 Caption: Weighted two-sample t-test used to calculate p-value.

Supplementary Figure 2



Supplementary Figure 2 Caption: Estimated average percentage of people in different age groups (under 30, between 30 and 44, between 45 and 64, 65 and older) who were exposed to at least one hard news website (N = 3,676). The panel on the left represents 2016 and the panel on the right represents 2020. Higher values indicate a greater estimated percentage of people in each age group who were exposed to at least one hard news website. Data are presented as weighted mean values $\pm$ 95% confidence intervals.

Supplementary Table 6

	Exposure to at least one hard news website (binary)	
	2016	2020
Trump supporter	0.002 (0.013)	0.016 (0.019)
Political knowledge	0.005 (0.003)	0.022* (0.008)
Political interest	0.015 (0.008)	0.010 (0.011)
College	0.024 (0.013)	0.048* (0.020)
Female	0.006 (0.012)	0.018 (0.018)
Non-white	0.006 (0.013)	0.009 (0.021)
Age 30-44 years	0.010 (0.019)	0.071* (0.030)
Age 45-64 years	0.009 (0.017)	0.088** (0.028)
Age 65+	-0.008 (0.020)	0.119*** (0.031)
Constant	0.826*** (0.027)	0.701*** (0.043)
Observations	2,514	1,151
R ²	0.009	0.034
Adjusted R ²	0.005	0.027
Residual Std. Error	0.276 (df = 2504)	0.303 (df = 1141)
F Statistic	2.489** (df = 9; 2504)	4.486*** (df = 9; 1141)

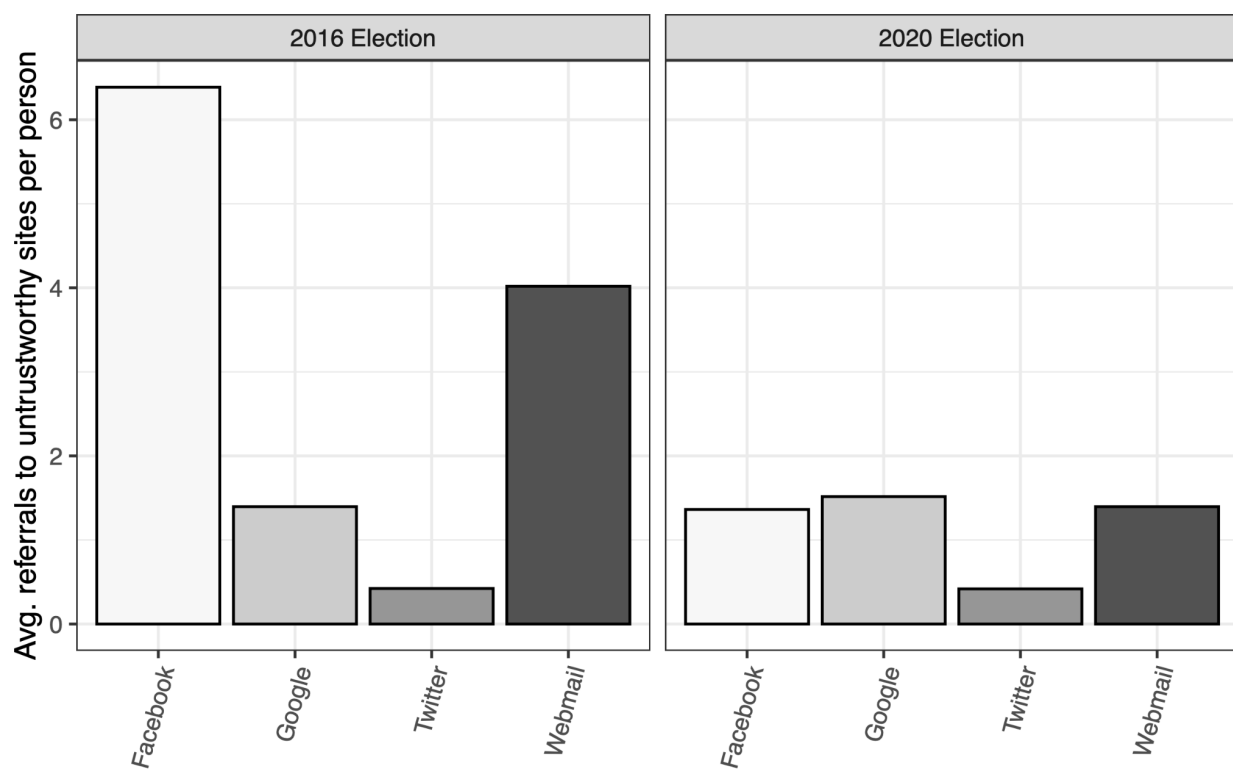
Note:

*p<0.05; **p<0.01; ***p<0.001

Supplementary Table 6 Caption: OLS regression coefficients are shown with standard errors in parentheses (models estimated using survey weights). The model on the left represents the 2016 election and the model on the right represents the 2020 election. The dependent variable in both models is a binary variable indicating whether an individual was exposed to at least one hard news website during the data collection period (1) or not (0). “Trump supporter” variable: intending to vote for Trump in 2020 election = 1; not intending to vote for Trump in 2020 election = 0. “Political knowledge” variable: variable ranging from 0-4 representing the number of questions in Pew Research Center’s civic knowledge questionnaire answered correctly out of 4. “Political interest” variable: variable ranging from 1-4 where 4 = people who say they pay attention to what’s going on in government and politics “most of the time” and 1 = those who pay attention “hardly at all”. “College” variable: 1 = college graduate; 0 = not a college graduate. “Female” variable: 1 = indicated identifying as a female, 0 = did not indicate identifying as a female. “Non-white” variable: 1 = indicated identifying as a race other than white; 0 = indicated identifying as white. P-values are two-sided and not corrected for multiple comparisons.

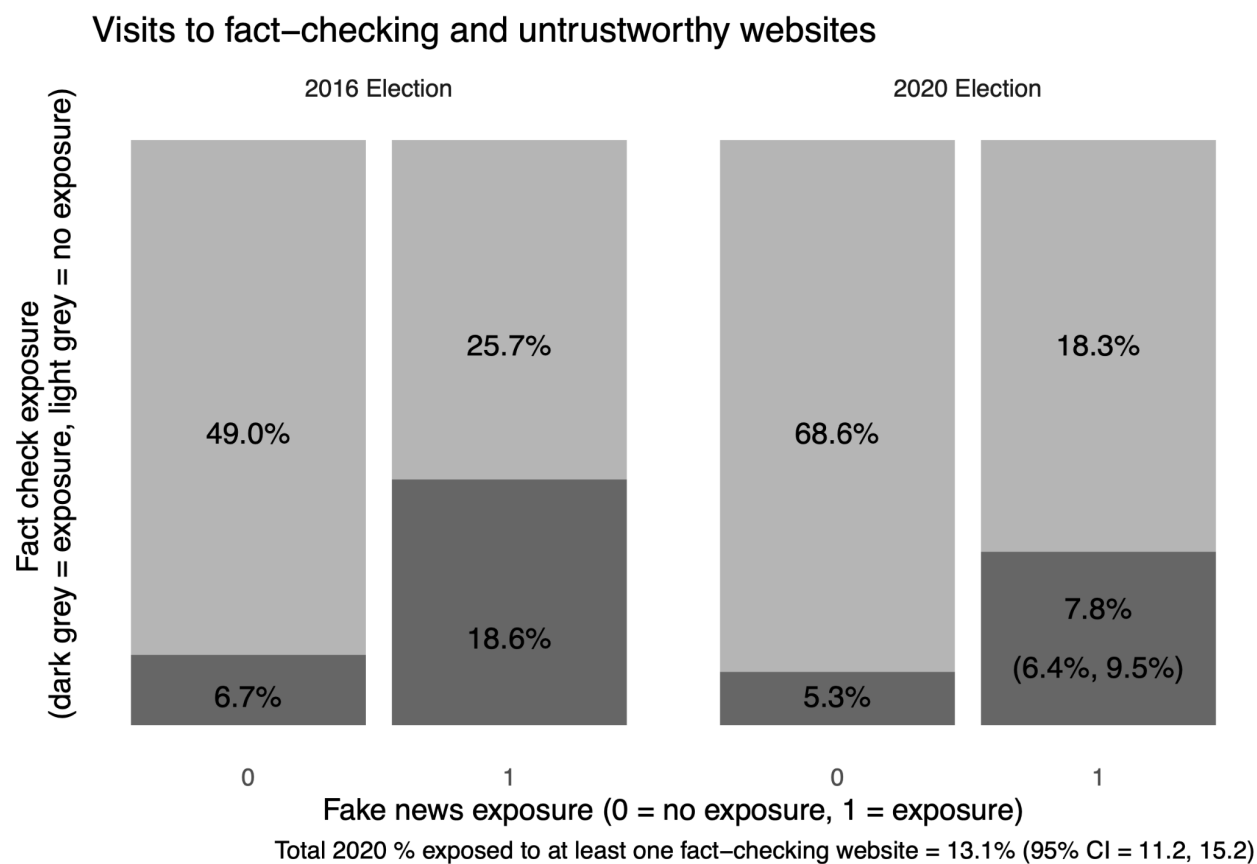
Platforms' average referrals to untrustworthy websites per exposed person

Supplementary Figure 3: Average referrals to untrustworthy websites per person exposed to at least one untrustworthy website



Fact-checking exposure

Supplementary Figure 4



Supplementary Figure 4 Caption: Per Guess et al. (2020), fact check exposure was measured as a visit to PolitiFact, the Washington Post Fact Checker, Factcheck.org, or Snopes.

Additional information for media slant deciles

Supplementary Table 7: Additional information for media slant decile figure (Figure 1 in the Main Paper)

decile	2016				2020			
	M	SE	min	max	M	SE	min	max
1	0.50	0.06	0	922	0.17	0.04	0	275
2	0.54	0.06	0	332	0.34	0.07	0	21
3	0.57	0.05	0	258	0.22	0.05	0	10
4	0.45	0.06	0	148	0.15	0.05	0	57
5	0.43	0.05	0	616	0.22	0.05	0	50
6	0.42	0.06	0	445	0.21	0.07	0	62
7	0.33	0.05	0	276	0.22	0.06	0	32
8	0.24	0.04	0	236	0.30	0.07	0	41
9	0.61	0.06	0	338	0.56	0.07	0	207
10	0.85	0.04	0	1554	0.76	0.06	0	912

Topline 2020 results using Guess et al. (2020)'s untrustworthy website list

Below are the findings from the Results section of the main paper but using the list of 490 untrustworthy websites from Guess et al. (2020) rather than the larger and more up-to-date list of 1,796 untrustworthy websites we constructed as described in the Methods section. Guess et al. (2020) compiled their list of 490 sites from databases of untrustworthy websites created by Allcott & Gentzkow (2017) and Grinberg et al. (2019).

Total Consumption of Untrustworthy Websites in 2020

During the 2016 election, Guess et al. (2020) found that 44.3% (95% CI = 40.8-47.7%) of Americans aged 18 or older were exposed to at least one untrustworthy website. By contrast, we find that during the 2020 election only 19.2% (95% CI = 15.9-22.5%) of Americans were exposed to at least one untrustworthy website. This decrease represents a significant reduction ($t = -16.44$, $p < .001$) in the percentage of Americans exposed to untrustworthy websites from 2016 to 2020.

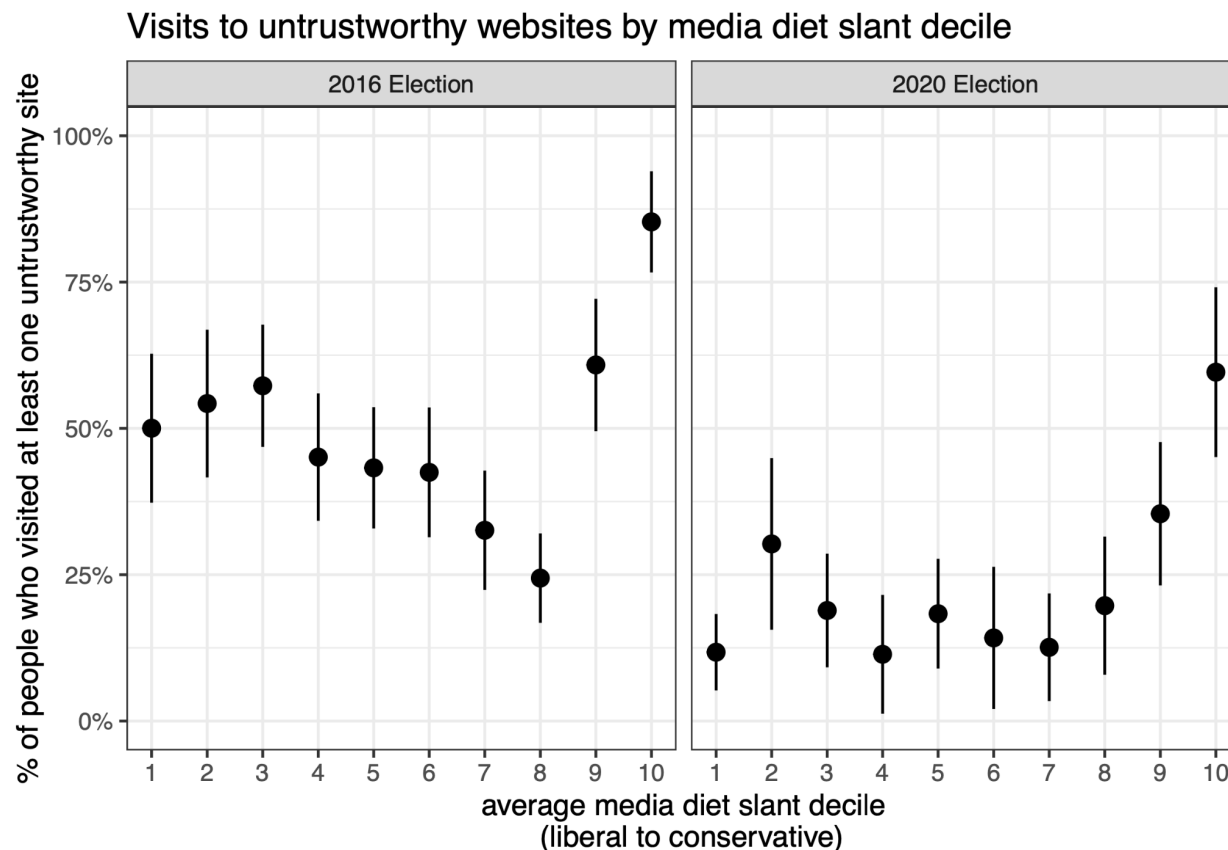
While a smaller percentage of Americans appeared to be exposed to any untrustworthy websites in 2020 compared to 2016, one possibility is that those who were exposed to untrustworthy websites in 2020 were exposed at greater rates than in 2016. However, our data suggest otherwise. First, the average number of untrustworthy website visits among those who visited at least one untrustworthy website in 2020 (15.6 visits; 95% CI = 9.43-21.8) was significantly lower than the average number in 2016 (32.1 visits; 95% CI = 26.3-37.9) ($t = -3.82$, $p < .001$). Second, the average length of time spent on untrustworthy websites was shorter in 2020 (32.0 seconds; 95% CI = 26.6-37.3s) than in 2016 (64.2 seconds).

Who Consumed Untrustworthy Websites in 2020?

In 2016, Guess et al. (2020) found that supporters of presidential candidate Donald Trump were significantly more likely to visit untrustworthy websites than supporters of his opponent Hillary Clinton, with 59.5% (95% CI = 54.5-64.5%) of Trump supporters visiting at least one untrustworthy website and 37.1% (95% CI = 31.9-42.2%) of Clinton supporters visiting at least one untrustworthy website. We find a similar difference between Donald Trump and Joe Biden supporters in 2020. 27.4% (95% CI = 21.2-33.3%) of Trump supporters visited at least one untrustworthy website in 2020 while 13.1% (95% CI = 9.2-17.1%) of Biden supporters visited at least one untrustworthy website. However, for both Trump and Clinton/Biden

supporters, the percentages of supporters exposed in 2020 were significantly lower than the percentages of supporters exposed in 2016 ($t_{\text{Trump}} = -11.9, p < .001$; $t_{\text{Clinton/Biden}} = -12.6, p < .001$).

Supplementary Figure 5 (Guess et al. 2020 untrustworthy website list)

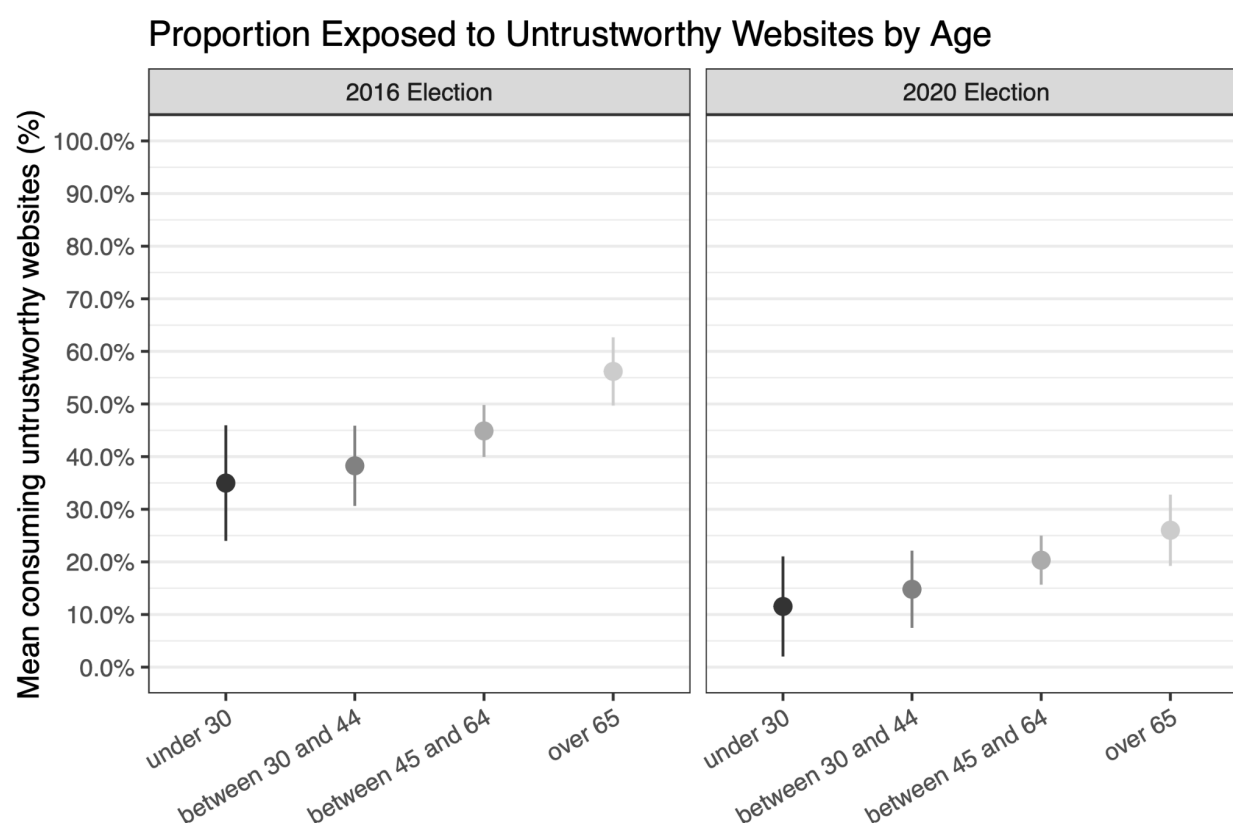


Supplementary Figure 5 Caption: Estimated percentage of people in media diets with different ideological slants who were exposed to at least one untrustworthy website contained in the Guess et al. (2020) untrustworthy website list ($N = 3,432$). The panel on the left represents 2016 and the panel on the right represents 2020. Individuals' media diets were split into deciles based on the ideological slant of their news consumption as determined by ideological ratings of hard news websites from Bakshy et al. (2015). Along the x-axis, lower values represent more liberal media diets, middle values represent more moderate media diets, and higher values represent more conservative media diets. Along the y-axis, higher values indicate a greater estimated percentage of people in that media slant decile who were exposed to at least one untrustworthy website. Guess et al. (2020) calculated 2016 deciles separately for Trump and Clinton supporters; we re-calculated them to be among all participants using their publicly available replication data. Data are presented as weighted mean values $\pm$ 95% confidence intervals.

Age also played an important role in exposure to untrustworthy websites during the 2016 election, with older individuals generally being more likely to visit untrustworthy sites. In 2016, those 65 and older were 1.6 times more likely to visit untrustworthy websites than adults under 30. During the 2020 election, it appears that the positive association between age and likelihood

of exposure to untrustworthy websites persisted, with those 65 and older being 2.3 times more likely to visit untrustworthy websites than those under 30. However, all age groups in 2020 were less likely to be exposed to untrustworthy websites than those same age groups in 2016 (see Figure S6). For instance, while 56.2% (95% CI = 49.7-62.7%) of people 65 and older were exposed to untrustworthy websites in 2016, only 26% (95% CI = 19.2-32.8%) of people 65 and older were exposed in 2020.

Supplementary Figure 6 (Guess et al. 2020 untrustworthy website list)



Supplementary Figure 6 Caption: Estimated average percentage of people in different age groups (under 30, between 30 and 44, between 45 and 64, 65 and older) who were exposed to at least one untrustworthy website contained in the Guess et al. (2020) untrustworthy website list (N = 3,676). The panel on the left represents 2016 and the panel on the right represents 2020. Higher values indicate a greater estimated percentage of people in each age group who were exposed to at least one untrustworthy website. Data are presented as weighted mean values +/- 95% confidence intervals. Lighter colors correspond to older age groups.

To examine the association between a variety of individual differences and exposure to untrustworthy websites, we follow the strategy used by Guess et al. (2020) and estimate a linear

probability model for the 2020 election period. In this model, the dependent variable is a binary variable indicating whether an individual was (1) or was not (0) exposed to at least one untrustworthy website during the data collection period. The independent variables include presidential candidate support, political knowledge, political interest, level of education, gender, race, and age. The models for 2016 and 2020 are presented in Table S8. Across both 2016 and 2020, many of the same demographic variables were significantly associated with exposure to untrustworthy websites. Specifically, being a Trump supporter, possessing greater political knowledge, and being more interested in politics were associated with a significantly greater likelihood of being exposed to at least one untrustworthy website in both 2016 and 2020 (all p 's < .05).

Supplementary Table 8 (Guess et al. 2020 untrustworthy website list)

	Exposure to at least one untrustworthy website (binary)	
	2016	2020
Trump supporter	0.164*** (0.021)	0.121*** (0.024)
Political knowledge	0.017** (0.005)	0.056*** (0.011)
Political interest	0.048*** (0.014)	0.026* (0.013)
College	0.040 (0.022)	0.025 (0.025)
Female	-0.022 (0.020)	0.006 (0.023)
Non-white	-0.094*** (0.021)	-0.020 (0.026)
Age 30-44 years	0.041 (0.032)	0.024 (0.038)
Age 45-64 years	0.064* (0.029)	0.048 (0.035)
Age 65+	0.129*** (0.033)	0.073 (0.039)
Constant	0.126** (0.046)	-0.117* (0.054)
Observations	2,514	1,151
R ²	0.092	0.083
Adjusted R ²	0.089	0.076
Residual Std. Error	0.466 (df = 2504)	0.382 (df = 1141)
F Statistic	28.265*** (df = 9; 2504)	11.486*** (df = 9; 1141)

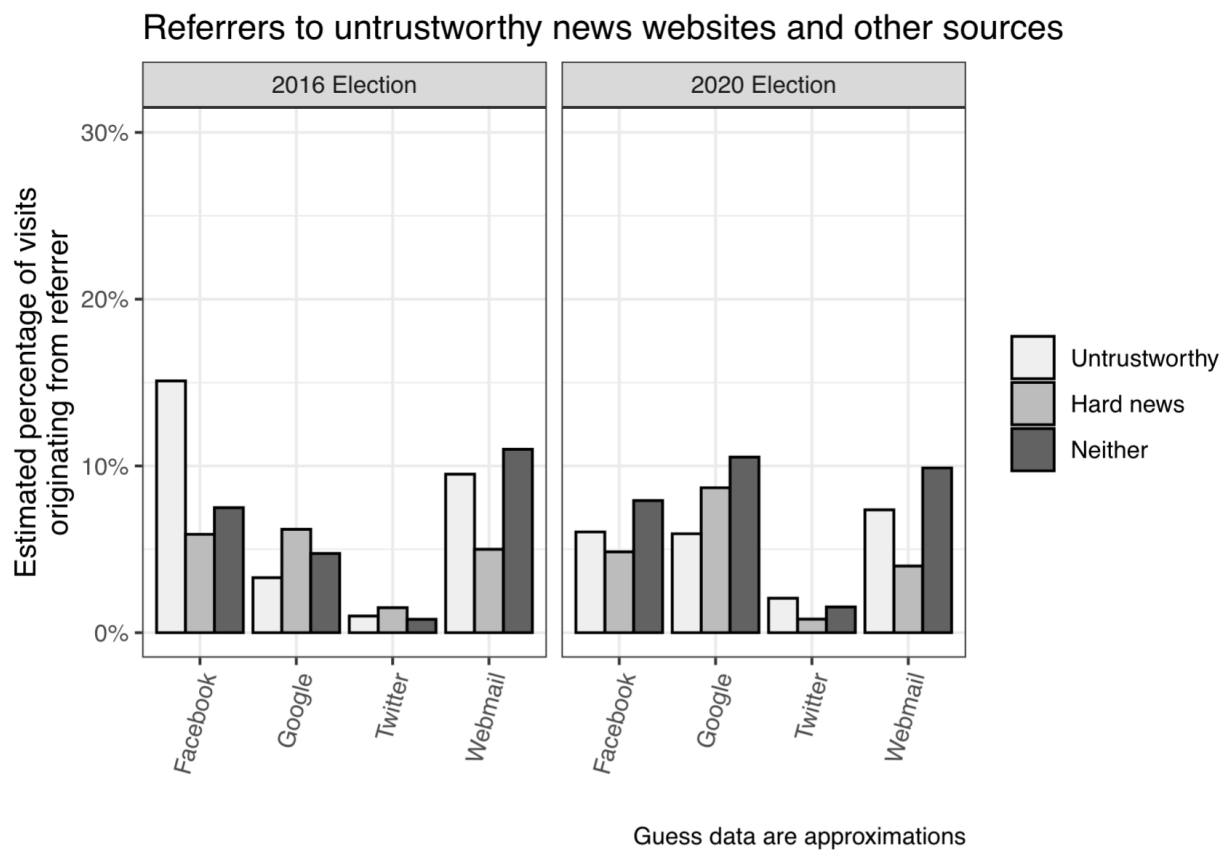
Note:

*p<0.05; **p<0.01; ***p<0.001

Supplementary Table 8 Caption: OLS regression coefficients are shown with standard errors in parentheses (models estimated using survey weights). The model on the left represents the 2016 election and the model on the right represents the 2020 election. The dependent variable in both models is a binary variable indicating whether an individual was exposed to at least one untrustworthy news website contained in the Guess et al. (2020) untrustworthy website list during the data collection period (1) or not (0). “Trump supporter” variable: intending to vote for Trump in 2020 election = 1; not intending to vote for Trump in 2020 election = 0. “Political knowledge” variable: variable ranging from 0-4 representing the number of questions in Pew Research Center’s civic knowledge questionnaire answered correctly out of 4. “Political interest” variable: variable ranging from 1-4 where 4 = people who say they pay attention to what’s going on in government and politics “most of the time” and 1 = those who pay attention “hardly at all”. “College” variable: 1 = college graduate; 0 = not a college graduate. “Female” variable: 1 = indicated identifying as a female, 0 = did not indicate identifying as a female. “Non-white” variable: 1 = indicated identifying as a race other than white; 0 = indicated identifying as white. P-values are two-sided and not corrected for multiple comparisons.

How are People Exposed to Untrustworthy Websites in 2020?

Supplementary Figure 7 (Guess et al. 2020 untrustworthy website list)



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